ÇANKAYA UNIVERSITY
Software Engineering Department
SENG 491 – 492 Project Proposal Form

Part I. Project Proposer

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Part II. Project Information. To be completed by the Proposer (Supervisor, Student or Company)

Starting Term	2024 / 2025	○ Fall	○ Spring
Title of the Project	AI Driven Pet Disease-Diagnosis-Treatment Manager Tool for both Veterinarians and Pet Owners (Pet Care SaaS Platform)		

Project Description

This project is a multiplatform SaaS solution designed to streamline interactions between pet owners and veterinarians. It features both web and mobile applications built in .NET, offering various tools to improve pet health management.
Pet owners can submit health-related media (e.g., photos, videos) and fill out AI-integrated voice surveys to receive a preliminary diagnosis for their pet's condition. The platform also enables users to compare their pet's health data with other animals that received the same official diagnosis from veterinarians, helping them make informed decisions.

Additionally, the platform offers features such as price comparisons for treatments, a map of nearby veterinarians within a 10 km radius, appointment booking, pet-friendly taxi and shop services, and payment processing (via the app or in cash). When an appointment is made, veterinarians receive notifications with attached medical surveys and AI diagnoses for review before the visit.

Project Justification

Novelty/Innovation	
Previous work	There are existing platforms for veterinarian appointments (such as PetDesk or Vetstoria), but none that combine media-based AI pre-diagnosis, health survey comparison, cost estimates, and additional services like pet taxis and shops, all within one ecosystem.
New aspects	This project introduces the integration of AI for media-based diagnosis and the comparison of health surveys among pet owners whose pets share similar medical conditions. The platform also connects additional pet services such as transportation and shopping into the vet-care process, enhancing user convenience.
Complexity	
Challenging problem and issues	• Data Privacy: Handling sensitive data (medical surveys, payment information) securely while maintaining regulatory compliance is crucial. • Multiplatform Development: Designing and maintaining consistent functionality across both web and mobile platforms requires robust development and testing. • Real-time Synchronization: Managing notifications and real-time updates for veterinarians and pet owners (e.g., appointment details, AI diagnosis) while ensuring low latency will require an efficient backend system. • AI Accuracy: Ensuring the AI can make accurate pre-diagnoses based on media and health surveys poses a technical challenge.
Related computer science fields and subfields	• Multiplatform Application Development (Web and Mobile) • Health Informatics • Payment Gateway Integration • Cybersecurity and Data Privacy • AI and Machine Learning (for diagnosis and prediction) • Cloud-Based Services and Scalability

References to existing solutions (methods, algorithms)	• Natural Language Processing (NLP): For AI-based voice surveys • Geolocation Services: To locate nearby veterinarians • Payment Integration: Stripe, PayPal, or similar payment gateways • Notification Systems: Firebase or SignalR for real-time communication • Machine Learning: Convolutional Neural Networks (CNNs) for image recognition (diagnosis via media)
Tools and Libraries	• AI/ML Libraries: TensorFlow, PyTorch • Cloud Infrastructure: Microsoft Azure, AWS • Database: SQL Server, Firebase Realtime Database • Geolocation APIs: Google Maps API, Here API • Payment Gateways: Stripe, PayPal APIs • Security Libraries: JWT for authentication, OAuth2 • .NET (Blazor for web), Python (AI/ML modules), React native for mobile
Proposed Team Size	4 People
Risk involved	
Potential problems and alternative solutions	• Data Breaches: Sensitive data must be protected through encryption and security best practices. Regular audits and vulnerability testing will be part of the development process. • Platform Incompatibility: Ensuring consistent performance across both web and mobile platforms may require additional resources. This risk can be minimized through rigorous testing and the use of shared codebases (e.g., .NET MAUI). • AI Misdiagnosis: If AI misdiagnoses, this could lead to legal or trust issues. We will mitigate this by emphasizing the preliminary nature of the AI diagnosis and ensuring veterinarians provide final diagnoses.
Minimum work required	The project requires a minimum of 4 people, working 8 hours per week over two semesters. Tasks will include AI model training, frontend and backend development for both web and mobile platforms, and integration of third-party services.
Evaluation	
Evaluation method (dataset, usability tests, metrics, etc.)	• Usability Testing: Focus groups and surveys from pet owners and veterinarians • Performance Metrics: Real-time response times, platform uptime, and bug tracking • Security Testing: Penetration tests and compliance audits • User Retention and Satisfaction Surveys • AI Diagnostic Accuracy: Tested through controlled datasets and real-world user feedback